RESEARCH MINI-PROJECT

Can gene expression predict cancer-drug sensitivity?

649

matched cell lines

286

drug-name groups

160,512

observed response pairs

Question: What generalizes to new cells, and what generalizes to new drugs?

What is being predicted?

Cell profile

Gene expression
Mutation records
Gene copy number

Drug information

Compound identity if known
Putative targets
Pathway labels

Response

Continuous LN_IC50
Fitted laboratory IC50
Lower = less concentration

A prediction is one cell-line and one drug combination.

DATASET

How did we prepare the data?

649

Molecular profiles

cross-omics matched cells

286

Drug groups

source-normalized names

160,512

Observed responses

86.5% of matched grid

No missing response values were invented. Only verified, available measurements were modeled.

The prediction pipeline

01

Match sources

COSMIC / DepMap IDs

02

Hold out entities

Cells or drugs

03

Build features

100 expression + 20 genomic + drug

04

Fit model

Training-only preprocessing, Ridge

05

Evaluate

Held-out RMSE, MAE, per-drug r

Predictor: scaled Ridge regression (alpha = 100); baseline models are fitted only to training responses.

Three held-out tests

New cell lines

Train: different cell lines

Test: cell IDs never used in training

Fair baseline: mean for each familiar drug

New drugs

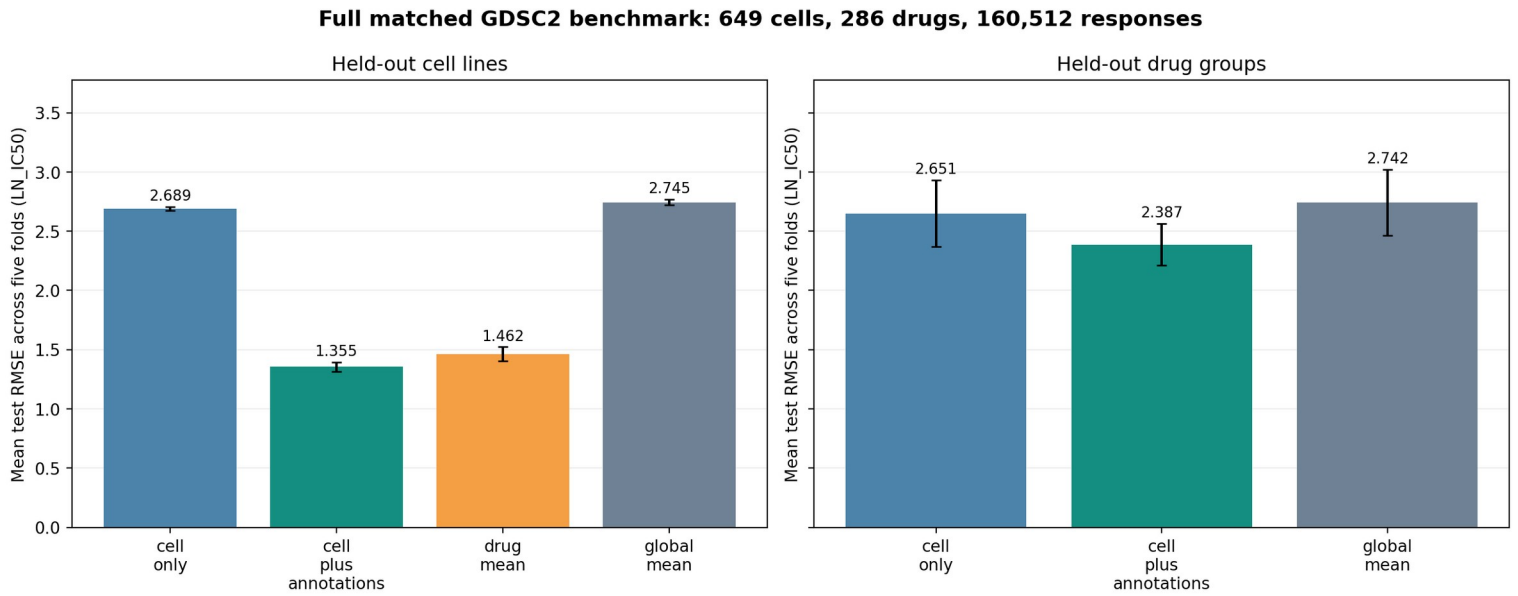
Original: disjoint drug-name groups

Revised: disjoint known PubChem CIDs

50 drug codes remain unverified

Revised BOTH NEW (32,149 pairs): RMSE 2.501 combined vs 2.554 annotations-only.

What did the model achieve?



1.355

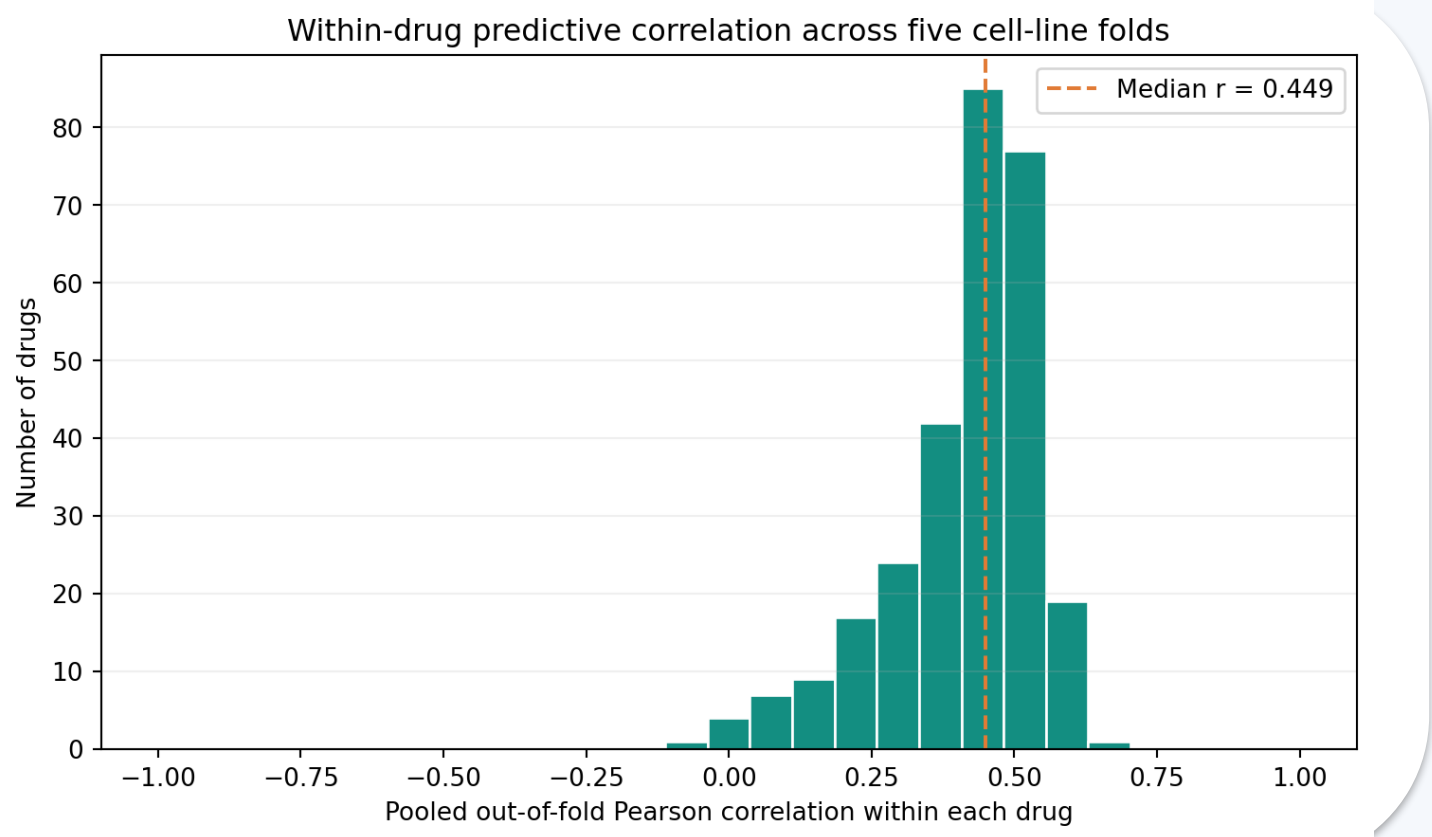
Unseen-cell RMSE
vs 1.462 baseline

2.452

CID-grouped drug RMSE
vs 2.550 annotation-only

Figure: original normalized-name folds. CID-grouped result shown at right.

Pooled correlation is not the whole story



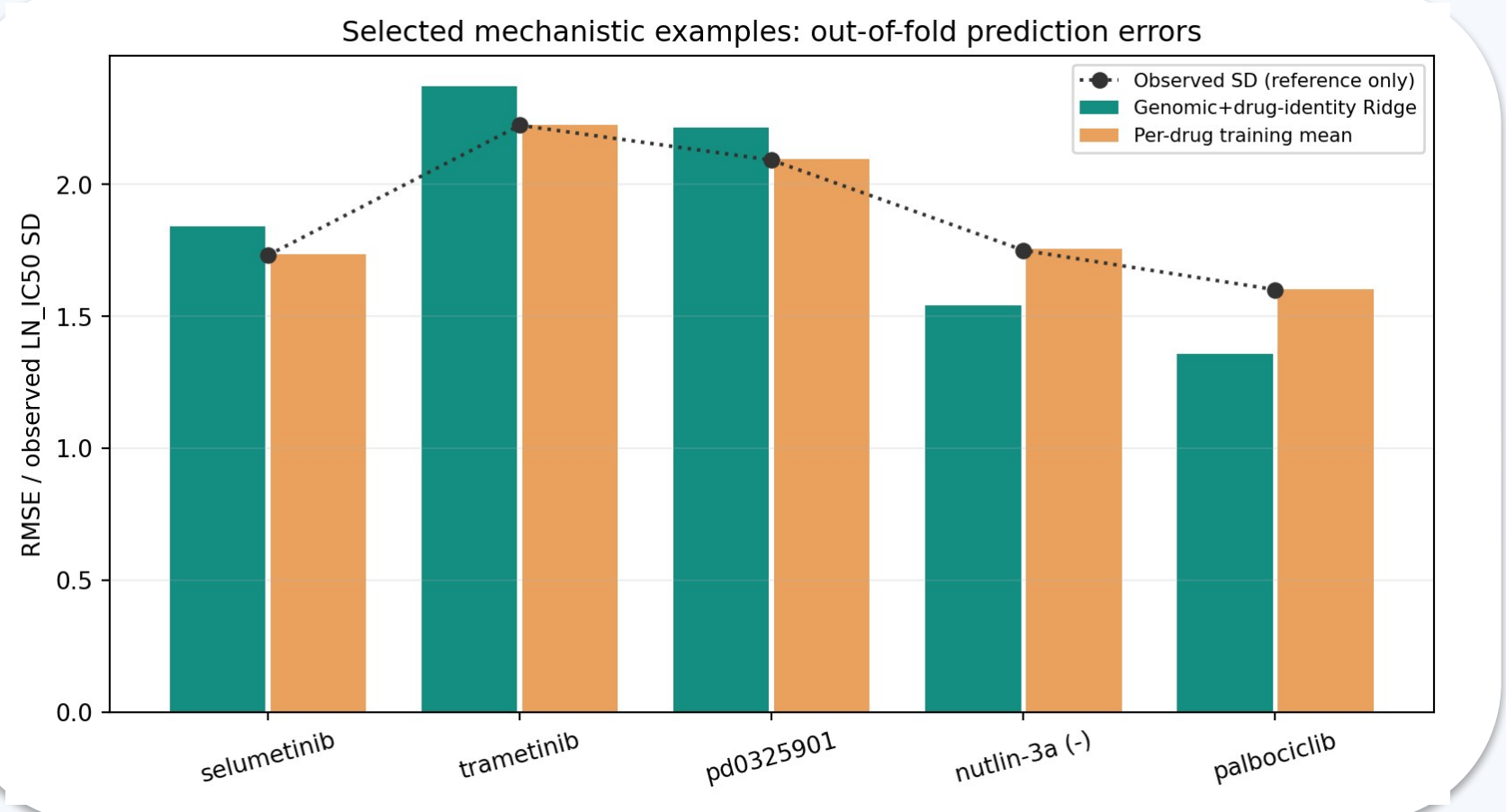
0.449

Median within-drug r

229/286

Drugs with lower RMSE than their training mean

Which drugs remain difficult?

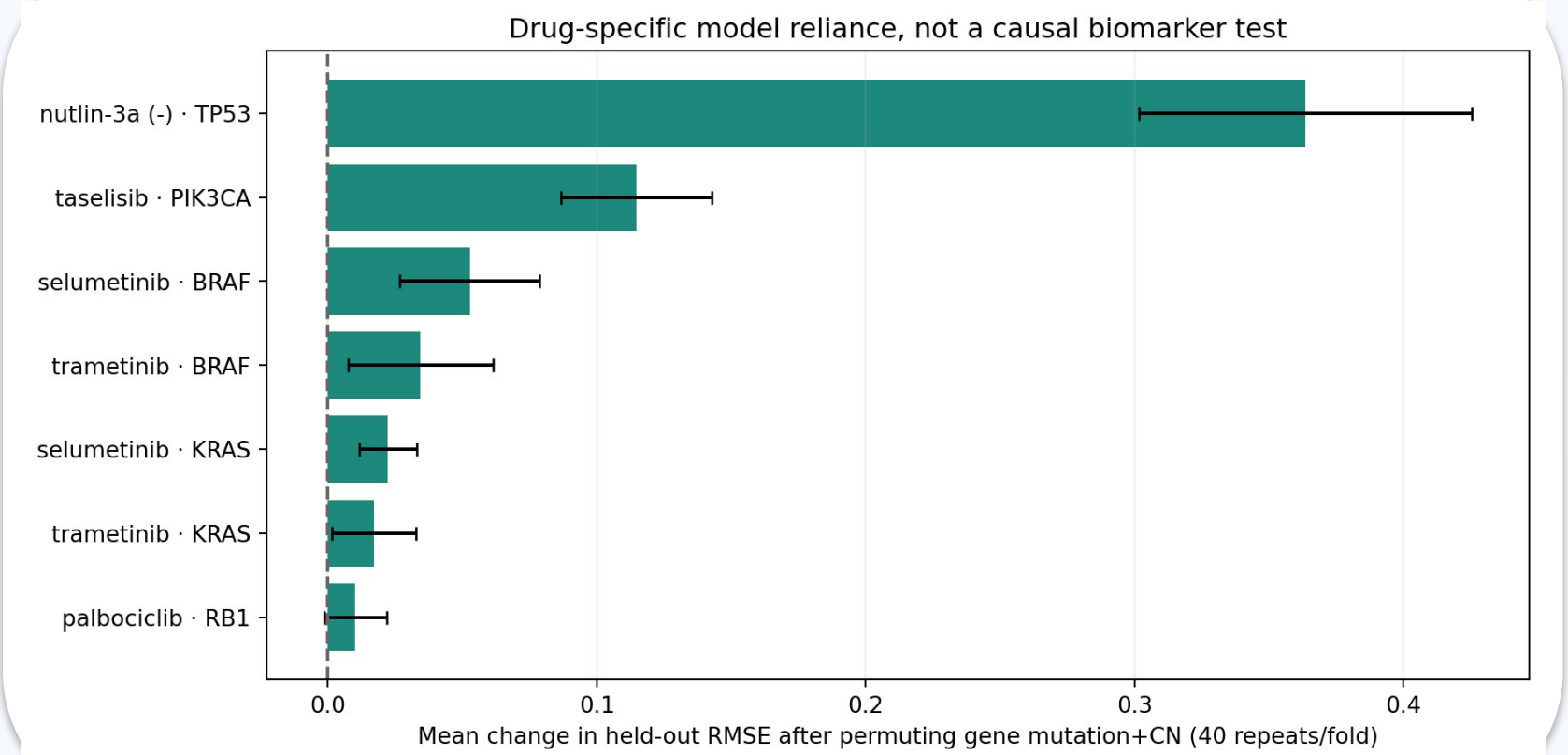


Key observation

A known drug identity does not ensure accurate variation between cell lines.

MEK inhibitors remain challenging.

Can genomic variables explain some responses?



Nutlin-3a / TP53

Mechanism-linked model reliance on held-out cells.

Biological association, not a validated treatment biomarker.

What can we conclude safely?

Finding

Better than appropriate baselines on both grouped holdout tasks; within-drug performance varies.

Assumptions

Cross-source cell IDs and supplied drug target/pathway annotations are meaningful.

Limitations & uncertainty

Scientific risks: 50 missing CIDs; 2 chemical-form conflicts; retrospective tests; unverified salt forms; source-year mismatch; no external screen.

Nonclinical research; incomplete chemical aliases; AI-assisted development disclosed.